

Exploring the Applicability of Digital Neuropsychological Assessments: A Comparative Study of CANTAB and CERAD-K in Elderly Koreans

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Abstract

This study examines the potential applicability of the Cambridge Neuropsychological Test Automated Battery (CANTAB) for cognitive assessment in elderly Koreans, compared to the Consortium to Establish a Registry for Alzheimer's Disease Korean version (CERAD-K). Factor analyses identified distinct cognitive domains in both tools, with CANTAB assessing broader constructs such as spatial working memory and switching/inhibition, while CERAD-K emphasized traditional domains like verbal ability and episodic memory. Age and education significantly influenced performance across cognitive domains, with older and less-educated participants demonstrating poorer outcomes. Gender differences were minimal, except for processing speed. Memory-related CANTAB subtests correlated with all CERAD-K cognitive domains, while other CANTAB subtests displayed task-specific relationships. These findings suggest the complementary nature of CANTAB and CERAD-K, underscoring the need for culturally tailored norms. However, the study's limitations, including a small sample size and cross-sectional design, necessitate cautious interpretation. Further large-scale, longitudinal research is needed to validate CANTAB's diagnostic potential for age-related cognitive decline in diverse Korean populations.

Keywords : Cognitive Assessment, Elderly Koreans, CANTAB, CERAD-K, Digital Neuropsychological Assessment

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1. Instruction

As global populations continue to age, the demand for refined cognitive assessment tools capable of identifying early-stage cognitive impairments, such as dementia and mild cognitive impairment (MCI), has become increasingly critical (Petersen et al., 2014). In Korea, where the elderly demographic is rapidly expanding, there is an urgent need for precise, reliable cognitive screening instruments to address the growing incidence of age-related cognitive decline (Jeon, 2013; Park, Reuter-Lorenz, 2009). The Korean version of the Consortium to Establish a Registry for Alzheimer's Disease neuropsychological battery (CERAD-K) has been extensively utilized in clinical settings for evaluating cognitive decline associated with Alzheimer's disease and MCI (Lee et al., 2002). CERAD-K provides robust assessments across critical domains—episodic memory, language, and praxis—making it a cornerstone in dementia diagnosis among older Korean adults (Lee et al., 2002; Seo et al., 2010). However, CERAD-K's manual administration process can be time-intensive and may introduce variability in scoring, emphasizing the need for complementary tools that enhance both objectivity and efficiency in cognitive assessment (Müller et al., 2019).

Computerized neuropsychological batteries, particularly the Cambridge Neuropsychological Test Automated Battery (CANTAB), have therefore gained significant attention. CANTAB offers several advantages over traditional assessments, including automated scoring, standardized administration, and heightened sensitivity across domains such as memory, attention, and executive function (Barnett et al., 2015; Gonçalves, Pinho, Simões, 2017). The Paired Associates Learning (PAL) subtest within CANTAB, for example, has demonstrated efficacy in detecting early memory deficits linked to Alzheimer's disease and shows moderate correlation with conventional memory tests such as the Rey Complex Figure Test (Égerházi et al., 2007; Gonçalves, Pinho, Simões, 2017). Yet, Lenehan et al. (2016) found that CANTAB subtests do not consistently align with traditional cognitive domains. Their factor analysis of healthy older adults revealed that, although PAL was associated with episodic memory, other subtests, such as Spatial Working Memory (SWM) and Reaction Time (RTI), did not correlate strongly with discrete cognitive factors, indicating that CANTAB might assess broader constructs beyond the isolated domains typically captured in traditional assessments. This distinction underscores the need to explore CANTAB's potential as a complementary tool to traditional instruments like CERAD-K in building a comprehensive cognitive profile.

Research from Western contexts further indicates that CANTAB may assess unique cognitive domains not fully captured by traditional neuropsychological tests. For instance, Smith et al. (2013) found that CANTAB's working memory and spatial processing subtests displayed weak correlations with paper-based tests, suggesting divergence in the underlying constructs measured. Simpson et al. (2005) and Dassanayake, Ariyasinghe (2019) also highlighted that CANTAB scores are significantly influenced by demographic variables such as age, gender,

education, and cultural background, suggesting the necessity of tailored norms to enhance the accuracy of assessments across diverse populations. Specifically, age-related declines in processing speed, memory, and attention (Salthouse, 2019; Seidler et al., 2010) and gender differences in spatial memory and reaction time (Alexander, DeLong, Strick, 1986; Cools, D'Esposito, 2011) underscore the importance of demographic adjustments for precise cognitive evaluation. Furthermore, education-adjusted norms are essential to reduce biases in assessments, particularly for individuals with lower educational attainment, as higher education is frequently associated with greater cognitive resilience (Lenehan et al., 2016; Bento-Torres et al., 2017).

In the Korean context, integrating CANTAB with CERAD-K is especially relevant, given the unique educational and cognitive characteristics of the elderly population, where lower educational levels are common and may affect cognitive performance (Bento-Torres et al., 2017; Kim et al., 2016). Korean studies have consistently shown that sociodemographic variables significantly influence CERAD-K performance, particularly on subtests such as the Boston Naming Test and Word List Memory Test, further emphasizing the need for culturally adapted norms (Jang, Kim, 2012; Kim, et al., 2016). Additionally, recent work by Na et al. (2023) demonstrated strong correlations between CERAD-K and a tablet-based cognitive assessment, the Inbrain CST, suggesting that an integrated approach of digital and traditional tools could offer a more nuanced cognitive profile.

While digital devices often pose a significant challenge for elderly users, particularly those with limited experience or lower educational levels, their adoption in clinical settings is becoming increasingly unavoidable due to advancements in healthcare technology. Recognizing these challenges, this study incorporates a preliminary analysis of CANTAB and CERAD-K scores in a sample of 83 cognitively normal Korean elderly individuals. By examining the alignment of CANTAB subtests with CERAD-K across domains such as memory and attention, and assessing the influence of demographic variables (age, gender, education), this research seeks to evaluate the validity and complementarity of these tools. The findings are anticipated to contribute to the development of culturally relevant, cost-effective digital cognitive assessment tools tailored to the needs of the Korean elderly population.

2. Methods

2.1. Participants

The present investigation encompassed a cohort of 93 older adults aged 60 years and above. Participants were recruited through local community senior facilities, including senior welfare centers, and were limited to cognitively intact individuals with no antecedent history of neurological disorders, cranial trauma, or psychiatric diagnoses. To ensure cognitive integrity, only individuals whose Mini-Mental State Examination (MMSE) scores fell within normative parameters (Han et al., 2010), adjusted for gender, age, and educational attainment, were

included in the final analytical sample. This selection process yielded a dataset comprising 83 participants (57 females, 68.7%) ranging in age from 63 to 93 years ($M = 75.25$ years, $SD = 6.19$). The study protocol was approved by the Institutional Review Board of XX University and adhered to established ethical guidelines.

2.2. Procedure

All experimental procedures and assessments were conducted in a controlled, quiet laboratory environment under the supervision of the experimenter. Participants attended two discrete sessions: one dedicated to the administration of the traditional neuropsychological battery and another for the computerized neuropsychological battery (CANTAB). Upon arrival, participants were provided with a comprehensive orientation to the study protocol and subsequently furnished written informed consent. Following the consent process, participants completed a series of preliminary questionnaires and underwent screening assessments, specifically the MMSE. The first cognitive assessment session involved the administration of a standardized neuropsychological battery, including the CERAD-K. Experimenters adhered to the standardized sequence of tasks within the battery while integrating additional cognitive assessments beyond its scope. The second session comprised the administration of the CANTAB battery, during which participants were seated in front of a securely mounted tablet PC. After an initial orientation to the procedure, participants initiated the tasks independently once they indicated readiness. The CANTAB battery included audio instructions for each task, facilitating autonomous task completion. If participants encountered difficulty understanding a task, the experimenter provided clarifications during the practice trials, maintaining objectivity by positioning themselves outside the participant's field of view. Each session lasted approximately 75 to 90 minutes. To mitigate potential order effects, the sequence of sessions was randomized for each participant. Subsequent sessions were scheduled 2 to 4 weeks apart to ensure adequate separation.

2.3. Measures

1) Traditional paper-and-pencil neuropsychological battery

The Traditional Paper-and-Pencil Neuropsychological Battery includes a total of 13 tests, assessing various cognitive domains. Ten of these tests are derived from the CERAD-K battery, a standardized tool with strong reliability and validity for evaluating cognitive functions such as memory, verbal ability, constructional ability, and processing speed. Four additional tests were added to further assess these domains.

(1) Memory

Three CERAD-K tests were used to evaluate memory. The Word List Memory Test

involved presenting participants with a list of 10 words for recall across three trials, with the total number of correct recalls serving as the outcome measure. In the Word List Recall Test, participants attempted to recall as many words as possible from the initial list after a 5-minute delay; a storage rate was also calculated based on performance. The Word List Recognition Test assessed recognition by asking participants to identify the original words from a list of old and new words, with correct identifications recorded as the outcome.

(2) Verbal ability

Verbal ability was measured using four tests. In the Categorical Verbal Fluency Test, participants generated as many words as possible within the category "animals" in 1 minute, with correct responses recorded. The Boston Naming Test (BNT) required participants to name items on 15 image cards, with correct responses documented. The Controlled Oral Word Association Test (COWAT) involved listing words starting with specific letters, excluding proper nouns, within a 60-second limit per trial (Kang et al., 2000). The Vocabulary subtest in Korean Wechsler Adult Intelligence Scale-IV (K-WAIS-IV) evaluated expressive language and vocabulary through word definitions, increasing in complexity (Hwang et al., 2012).

(3) Constructional ability

Three tests were used to assess constructional ability. The Constructional Praxis Test involved reproducing four geometric figures, with accuracy as the outcome. The Constructional Praxis Recall Test required participants to recreate the figures after a 5-minute delay, measuring recall accuracy. In the Constructional Praxis Recognition Test, participants identified previously viewed figures among new ones.

(4) Processing speed

The Trail Making Test (TMT) was used to measure processing speed through two components. In TMT-A, participants connected numbered circles in sequence to assess attention, sequencing, and visuospatial tracking, with time as the outcome. TMT-B required alternating between numbers and letters, assessing mental flexibility, with completion time recorded.

(5) Working memory

Working memory was evaluated with the Letter-Number Sequencing (LNS) subtest (K-WAIS-IV), in which participants ordered sequences of numbers and days of the week, rearranging numbers in ascending order followed by days. The outcome was the total correct responses up to the test's stopping rule.

2) Computerized neuropsychological assessment: CANTAB

The CANTAB is a computerized battery that uses non-verbal stimuli presented on a touch-screen interface, assessing various cognitive domains. Six tests from the CANTAB battery were administered.

(1) Cognitive flexibility

The Multitasking Test (MTT) in the CANTAB battery assessed cognitive flexibility by requiring participants to alternate between two learned rules across three trials. In the first trial, participants followed the 'direction' rule, pressing a button corresponding to the direction of an arrow (left or right). In the second trial, they applied the 'position' rule, pressing the button corresponding to the location of the arrow on the screen. In the third trial, the task alternated between the two rules, with the active rule displayed as text at the top of the screen. This design tested participants' ability to adapt to changing demands, with performance measured by the number of correct responses (MTTtc) and errors (MTTic).

(2) Motor and cognitive response speed

The 5-Choice Reaction Time (RTI) task measured participants' motor and cognitive response speeds. Participants responded to a visual cue appearing randomly in one of five locations by lifting their finger from a start button and touching the illuminated target. Two key outcomes were median reaction time of correct responses (RTIcmdrt) and total response errors (RTIfes), reflecting response speed and accuracy.

(3) Spatial working memory

The Spatial Working Memory (SWM) task evaluated spatial memory retention and manipulation. Participants searched for hidden tokens within boxes while avoiding revisits to previously searched locations. Three primary measures included between errors (SWMbe), total errors (SWMte), and strategy score (SWMs), indicating spatial memory and search efficiency.

(4) Visual episodic memory and learning

The Paired Associates Learning (PAL) task assessed participants' ability to remember the locations of unique patterns within boxes. As the task progressed, the number of boxes with patterns increased, adding difficulty. Key outcomes were PAL total errors adjusted (PALtea), reflecting cumulative errors with adjustments for incomplete attempts, and PAL first attempt memory score (PALfams), measuring correct recalls on the first attempt across trials.

(5) Visual pattern recognition

The Pattern Recognition Memory (PRM) task evaluated visual recognition abilities. In this task, participants were shown a set of visual patterns, which they were later asked to identify in both immediate and delayed (approximately 20-minute) recognition trials. Two primary measures were analyzed: PRM percent correct immediate (PRMpci), which assessed accuracy in immediate recognition, and PRM percent correct delayed (PRMpcd), reflecting accuracy in delayed recognition.

(6) Verbal recognition memory

The Verbal Recognition Memory (VRM) task focused on verbal memory encoding and recognition. Participants read a list of words aloud, completed free recall, and engaged in immediate and delayed (approximately 20-minute) recognition trials. Primary outcomes were accuracy rates for both immediate and delayed recognition, reflecting short- and long-term verbal memory performance.

2.4. Analysis

Data analyses were conducted using SPSS 29.0. Descriptive statistics, including sample size, means, and standard deviations, were computed to provide a basic overview of the demographic characteristics and key variables. To explore the underlying structure of cognitive functions assessed by traditional and CANTAB test batteries, principal component analysis (PCA) was performed. Factors were extracted using the direct oblimin rotation method, with retention based on eigenvalues greater than one. Consistent with established guidelines, item factor loadings of $\geq .4$ were considered practically significant.

Composite scores for each factor were calculated by averaging the standardized scores of the tests loading onto each factor, based on the factor analysis results. This approach aligns with methodologies used in previous research (Andrade, 2021), where composite scores are derived from standardized test scores associated with each identified factor. Subsequently, correlation analyses were conducted to examine the relationships between factor scores from traditional paper-based tests and CANTAB assessments. Additionally, independent samples t-tests were performed on CANTAB composite scores to assess differences based on gender, age, and education level.

Finally, a PCA with direct oblimin rotation was conducted on all tests to investigate the overall structure of cognitive functions across both test batteries. The use of direct oblimin rotation is advantageous in this context as it allows for the identification of correlated factors, providing a more accurate representation of the interrelationships among cognitive domains.

Table 1. Descriptive statistics of demographic characteristics and neuropsychological assessments

Background characteristics	<i>N</i> (%)	<i>M</i>	<i>SD</i>
Age, years	83	75.25	6.19
Gender female	57 (68.7)	-	-
Chronic disease yes	55 (66.3)		
Anxiety	83	3.61	2.98
Depression	83	6.37	4.17
Education			
None	4 (4.8)	-	-
Elementary school	15 (18.1)	-	-
Middle school	13 (15.7)	-	-
High school	28 (33.7)	-	-
College	21 (25.3)	-	-
Greater than college	2 (2.4)	-	-
Traditional test battery			
MMSE	80	26.89	1.84
Categorical Verbal Fluency	79	15.13	4.44
BNT	79	13.42	1.39
Word List memory	79	18.49	4.23
Word List free recall	79	5.95	2.27
Word List storage	79	76.57	22.84
Word List recognition	79	9.05	1.60
Constructional Praxis recall	79	7.32	3.01
Constructional Praxis recognition	79	3.65	0.70
TMT A	80	65.99	44.67
TMT B	58	155.45	78.65
COWAT	80	24.99	9.35
Vocabulary (raw)	79	28.04	10.09
LNS (raw)	78	14.58	4.50
CANTAB test battery			
VRMirtc	80	30.24	3.16
VRMdrtc	79	29.48	3.64
PRMpci	80	89.38	12.37
PRMpcd	79	76.48	16.33
RTIfmdrt	79	414.07	46.51
RTIfes	79	0.77	1.58
MTTtc	79	135.75	18.60
MTTtic	79	17.63	15.65
MTTlmd	79	832.75	124.23
PALtea	79	30.63	15.77
PALfams	79	8.18	3.89
SWMte	79	19.25	8.61
SWMbe	79	18.66	8.33
SWMs	79	9.52	1.59

3. Results

3.1. Background

Table 1 presents a comprehensive summary of the participants' demographic characteristics and neuropsychological assessment outcomes. The study cohort comprised cognitively intact older adults with no documented history of neurological or psychiatric disorders. Cognitive functioning, evaluated using standardized MMSE norms ($M = 26.89$, $SD = 1.84$), indicated preserved cognitive abilities across the sample. Educational attainment was predominantly at the high school or university level. Notably, over half of the participants reported at least one chronic medical condition, such as hypertension or diabetes. Anxiety and depression levels, as measured by the Hospital Anxiety and Depression Scale (HADS), were generally within normal limits across the sample.

3.2. Factor analysis of traditional neuropsychological battery

To evaluate the factorability of the traditional neuropsychological battery, several preliminary analyses were conducted. The Kaiser - Meyer - Olkin (KMO) measure of sampling adequacy yielded a value of .67, surpassing the recommended threshold of .60 (Hair et al., 1998). Bartlett's test of sphericity was statistically significant, $X^2(78) = 382.00$, $p < .001$, confirming the suitability of the correlation matrix for factor analysis.

The factor analysis of the traditional neuropsychological battery revealed four distinct cognitive domains. The first factor, labeled memory, accounted for the largest proportion of variance (34.91%) and encompassed a range of memory-related tasks including Word List storage, Word List free recall, Word List recognition, Constructional praxis recognition, Word List memory, and Constructional praxis recall. The second factor, identified as verbal ability, explained 15.13% of the variance and comprised the Boston Naming Test and Categorical verbal fluency. Verbal intelligence emerged as the third factor, accounting for 9.90% of the variance, and included the COWAT and Vocabulary subtests. The fourth factor, processing speed, contributed 9.02% of the variance and consisted of TMT A and B, as well as Letter-number Sequencing. This factor structure highlights the multifaceted nature of cognitive functions assessed by the traditional neuropsychological battery, with a particular emphasis on memory and language-related abilities. However, only the Constructional praxis test did not demonstrate adequate factor loadings on any of the extracted factors and was therefore excluded from the analysis.

3.3. Factor analysis of CANTAB

The factorability of the six CANTAB tests was assessed using standard criteria. The KMO measure of sampling adequacy was .61, exceeding the minimum threshold of .60 (Hair et al., 1998). Bartlett's test of sphericity was significant, $X^2(91) = 784.74$, $p < .001$,

confirming the adequacy of the correlation matrix for principal component analysis (PCA). A PCA was conducted to elucidate the underlying structure of the CANTAB test battery. Four factors were extracted based on eigenvalues greater than 1, collectively accounting for 68.56% of the total variance. The first factor, designated as memory, accounted for 30.51% of the variance and encompassed PALfams, PALtea, PRMpci, PRMpcd, VRMirtc, and VRMdrtc. The second factor, termed SWM, explained 16.61% of the variance and comprised SWM indices. The third factor, labeled switching/inhibition, contributed an additional 11.94% of the variance and included MTT total correct and incorrect responses. The final factor, denoted as processing speed, accounted for an additional 9.50% of the variance and incorporated RTIfmdrt, MTTlmd, and RTIfes.

Table 2. Factor analysis of traditional neuropsychological tests

	Memory	Verbal ability	Verbal intelligence	Processing speed
Word List storage	0.906	0.086	0.208	-0.190
Word List free recall	0.902	0.100	0.212	-0.355
Word List recognition	0.731	-0.221	0.299	-0.172
Constructional Praxis recognition	0.715	-0.317	-0.064	-0.091
Word List memory	0.660	0.175	0.024	-0.463
Constructional Praxis recall	0.614	-0.422	0.231	-0.605
BNT	0.109	-0.782	0.007	-0.190
Categorical Verbal Fluency	0.069	-0.759	0.414	-0.303
COWAT	0.079	-0.267	0.817	-0.047
Vocabulary (raw)	0.279	0.095	0.791	-0.310
TMT A	-0.098	0.129	-0.074	0.817
TMT B	-0.212	0.176	-0.255	0.802
LNS (raw)	0.491	-0.304	0.163	-0.678

Table 3. Factor analysis of CANTAB tests

	Memory	SWM	Swiching/Inhibition	Processing speed
PALfams	-0.824	-0.262	-0.309	-0.076
PALtea	0.796	0.364	0.351	0.066
PRMpci	-0.699	0.085	-0.026	-0.271
PRMpcd	-0.662	-0.025	-0.093	-0.343
VRMirtc	-0.659	-0.201	0.149	-0.590
VRMdrtc	-0.619	-0.110	0.187	-0.560
SWMbe	0.203	0.957	0.207	0.042
SWMte	0.163	0.944	0.200	0.031
SWMs	0.009	0.771	-0.059	0.017
MTTtc	-0.270	-0.175	-0.910	-0.334
MTTtic	0.253	0.099	0.883	0.172
RTIfmdrt	0.269	0.075	0.177	0.774
MTTlmd	0.075	0.061	0.286	0.686
RTIfes	0.359	-0.054	-0.063	0.563

1) Group comparisons based on gender, age, and educational attainment

To further investigate the derived CANTAB factor scores, independent samples t-tests were conducted to examine differences across gender, age, and educational attainment groups. Results are summarized below and detailed in Table 4. Significant gender differences were observed for processing speed, $t(77) = -2.00$, $p < .05$, with males ($M = -0.23$, $SD = 0.84$) demonstrating lower scores than females ($M = 0.11$, $SD = 0.66$). No significant gender differences were found for other factors. Participants were stratified into two age categories: those under 75 years old and those 75 years or older. Significant age differences were observed for memory ($t(78) = 4.60$, $p < .001$), switching/inhibition ($t(77) = 3.36$, $p < .01$), and processing speed ($t(77) = 2.95$, $p < .01$), with the relatively younger elderly group (under 75) consistently demonstrating better performance compared to the older elderly group (75 and above). Differences for SWM were not statistically significant, $t(63.90) = 1.74$, $p = .087$. Educational attainment was dichotomized as middle school or below and high school or above. A significant effect of education level was found for memory, $t(78) = -3.99$, $p < .001$, with participants holding higher educational degrees ($M = 0.22$, $SD = 0.61$) scoring higher than those with lower educational attainment ($M = -0.41$, $SD = 0.77$). Additionally, switching/inhibition also showed a significant effect of education level, $t(77) = -2.33$, $p < .05$, with participants with higher-educated participants scoring higher ($M = 0.18$, $SD = 0.99$) than their less-educated counterparts ($M = -0.33$, $SD = 0.84$). No significant differences were observed for processing speed or SWM across educational levels. Furthermore, Multivariate Analysis of Variance (MANOVA) analyses examining the interaction effects between gender and age, as well as gender and educational attainment, revealed no statistically significant interactions for any of the CANTAB factor scores.

3.4. Correlation analysis between traditional factors and CANTAB subtests

The correlation analysis between traditional cognitive factors and CANTAB subtests revealed a nuanced relationship between these assessment modalities. While certain CANTAB subtests demonstrated moderate alignment with traditional composite measures, the degree of convergence varied substantially across cognitive domains. This heterogeneity in correlations underscores the distinct yet complementary nature of these neuropsychological test batteries, suggesting that each modality captures unique aspects of cognitive functioning in elderly populations.

3.5. Integrated factor analysis of traditional neuropsychological subtests and CANTAB

To elucidate the underlying structure of cognitive functions across both assessment modalities, a PCA was conducted on the combined CANTAB and traditional neuropsychological test batteries. The KMO measure of sampling adequacy was .59, approaching the recommended threshold of .60, and Bartlett's test of sphericity yielded

significant results ($X^2(351) = 1011.62, p < .001$), confirming the suitability of the data for factor analysis. The PCA yielded a seven-factor solution, collectively accounting for 68.15% of the total variance. Factor loadings and their respective contributions to variance are detailed in Table 6. Notably, the emergent factor structure revealed minimal overlap between the traditional battery and CANTAB subtests, with the majority of tests from each battery loading independently onto distinct factors.

Table 4. T-test of CANTAB composite scores by gender, age, and educational attainment

Gender	Male			Female			<i>t</i>	<i>df</i>	<i>p</i>
	<i>N</i>	<i>M</i>	<i>SD</i>	<i>N</i>	<i>M</i>	<i>SD</i>			
Memory	26	-0.10	0.56	54	0.03	0.80	-0.88	68.18 ^a	.381
SWM	26	0.05	1.05	53	-0.02	0.82	0.33	77	.740
Switching/Inhibition	26	-0.24	1.22	53	0.12	0.81	-1.54	77	.127
Processing speed	26	-0.23	0.84	53	0.11	0.66	-2.00	77	.049
Age	Under 75 years			75 and above			<i>t</i>	<i>df</i>	<i>p</i>
	<i>N</i>	<i>M</i>	<i>SD</i>	<i>N</i>	<i>M</i>	<i>SD</i>			
Memory	38	0.34	0.66	42	-0.33	0.65	4.60	78	< .001
SWM	38	0.18	1.05	41	-0.17	0.70	1.74	63.90 ^a	.087
Switching/Inhibition	38	0.36	0.75	41	-0.33	1.04	3.36	77	.001
Processing speed	38	0.24	0.65	41	-0.22	0.74	2.95	77	.004
Educational attainment	Middle school or below			High school or above			<i>t</i>	<i>df</i>	<i>p</i>
	<i>N</i>	<i>M</i>	<i>SD</i>	<i>N</i>	<i>M</i>	<i>SD</i>			
Memory	29	-0.41	0.77	51	0.22	0.61	-3.99	78	< .001
SWM	28	-0.18	0.73	51	0.10	0.97	-1.48	69.68 ^a	.144
Switching/Inhibition	28	-0.33	0.84	51	0.18	0.99	-2.33	77	.023
Processing speed	28	-0.14	0.72	51	0.08	0.74	-1.24	77	.218

^a Welch's t-test were conducted due to the violation of Levene's test for equality of variances.

Table 5. Correlations between CANTAB tests and composite scores of traditional test factors

	Memory	Verbal ability	Verbal intelligence	Processing speed
PALfams	.48***	.38***	.23*	.46***
PALtea	-.48***	-.42***	-.31**	-.39***
PRMpci	.38***	.29*	.26*	.39***
PRMpcd	.21	.26*	.10	.34**
VRMirtc	.47***	.17	.14	.28*
VRMdrtc	.55***	.23*	.41***	.26*
SWMbe	-.17	-.18	-.19	-.34**
SWMte	-.13	-.16	-.19	-.31**
SWMs	-.15	-.29*	-.08	-.21
MTTtc	.35**	.10	.31**	.38***
MTTtic	-.41***	.00	-.26*	-.27*
RTIfmdrt	-.34**	-.10	-.23*	-.36**
MTTlmd	-.06	-.06	-.25*	-.33**
RTIfes	-.24*	.00	-.08	-.11

p* < .05, *p* < .01, ****p* < .001

Table 6. Factor analysis of combined CANTAB and traditional neuropsychological subtests

	1	2	3	4	5	6	7
Word List free recall	0.922	-0.024	-0.147	-0.133	0.174	0.203	-0.134
Word List storage	0.895	-0.021	-0.141	-0.103	0.147	0.115	-0.017
Word List memory	0.738	-0.034	-0.163	-0.128	0.041	0.257	-0.200
Constructional Praxis recall	0.600	-0.157	0.236	-0.283	0.311	0.362	-0.569
Word List recognition	0.584	-0.126	0.218	0.017	0.386	0.423	-0.078
Constructional Praxis recognition	0.529	-0.258	0.262	-0.049	-0.148	0.097	-0.312
VRMdrtc	0.445	0.004	0.334	-0.362	0.439	0.439	0.128
SWMbe	-0.048	0.956	0.021	-0.012	0.014	-0.149	0.173
SWMte	-0.017	0.944	0.024	-0.018	0.003	-0.107	0.163
SWMs	-0.188	0.759	-0.161	-0.066	-0.218	-0.148	0.013
MTTtc	0.149	-0.055	-0.869	-0.158	0.146	0.105	-0.182
MTTtic	-0.262	-0.006	0.857	0.042	-0.111	-0.104	0.150
RTIfes	-0.115	-0.084	-0.147	0.682	0.084	-0.185	-0.098
MTTlmd	0.063	-0.082	0.291	0.676	-0.173	0.038	0.061
RTIfmdrt	-0.119	-0.041	0.124	0.642	-0.024	0.087	0.435
VRMirtc	0.327	-0.132	0.241	-0.599	-0.054	0.538	0.146
COWAT	0.039	-0.097	-0.069	-0.089	0.765	-0.009	-0.059
Vocabulary (raw)	0.232	0.009	-0.231	0.022	0.680	-0.023	-0.180
Categorical Verbal Fluency	0.019	-0.097	0.168	0.137	0.588	0.375	-0.418
PALtea	-0.296	0.271	0.113	0.058	-0.181	-0.843	0.144
PALfams	0.268	-0.142	-0.076	-0.086	-0.004	0.837	-0.283
BNT	0.062	-0.228	0.326	0.019	0.321	0.536	-0.214
PRMpcd	0.193	0.141	-0.017	-0.469	-0.089	0.531	-0.242
PRMpci	0.202	0.397	0.210	-0.301	0.042	0.525	-0.202
TMT A	-0.054	0.133	0.136	0.102	-0.069	-0.185	0.820
TMT B	-0.316	0.088	0.056	-0.022	-0.310	-0.291	0.670
LNS (raw)	0.456	-0.244	0.035	-0.488	0.183	0.461	-0.559

4. Discussion

This study identified distinct cognitive factors through analyses of traditional paper-and-pencil tests and CANTAB subtests, showcasing the unique strengths and contributions of each methodology. Factor analysis of the paper-and-pencil tests revealed four primary cognitive domains: memory, verbal ability, verbal intelligence, and processing speed. Similarly, CANTAB subtests were grouped into factors reflecting memory, spatial working memory, switching/inhibition, and processing speed, highlighting the computerized tool's capacity to assess both broad and task-specific constructs. Within the elderly sample, age emerged as a significant variable, with older participants exhibiting greater deficits in memory and processing speed. Educational attainment also played a critical role; individuals with lower educational

levels performed worse in memory and switching/inhibition tasks. These effects of age and educational attainment align with previous findings that highlight their influence on overall cognitive functioning (Baek, Kim, 2022; Lee, Lee, 2019). Gender differences were minimal across most cognitive domains; however, women demonstrated superior performance in processing speed compared to men. This finding may be influenced by the age distribution within the sample, as female participants were generally younger ($M = 74.14$, $SD = 6.78$) than their male counterparts ($M = 77.69$, $SD = 5.62$), potentially confounding the relationship between gender and processing speed. Notably, memory-related CANTAB subtests, including PAL, PRM, and VRM, showed significant correlations with all cognitive domains assessed by paper-and-pencil tests, while other CANTAB subtests displayed task-specific relationships. These findings underscore the complementary nature of traditional and computerized cognitive assessments and emphasize the importance of demographic factors in interpreting cognitive performance among elderly populations.

The findings of this study align with and expand upon prior research in several important ways. Consistent with Smith et al. (2013), our results demonstrated low to moderate correlations between CANTAB subtests and paper-and-pencil composite scores. This suggests that while both approaches measure general cognitive factors, they capture distinct constructs that necessitate caution when assessing specific cognitive abilities. Similar to Lenehan et al. (2016), our factor analyses revealed that most cognitive factors were unique to either the paper-and-pencil or CANTAB subtests; notably, CANTAB exhibited a broader capacity to identify additional factors due to its diverse task inclusion.

While the integration of CANTAB and CERAD-K offers promising advances in cognitive assessment for elderly Koreans, digital implementation faces significant usability challenges. Addressing these barriers requires developing intuitive, age-appropriate interfaces and targeted training programs to enhance technological accessibility. Such strategic design considerations are crucial for effectively leveraging digital cognitive assessment technologies in geriatric clinical settings, ultimately aligning with emerging patient-centered healthcare innovations.

The memory-focused PAL subtest of CANTAB showed the strongest correlation with traditional memory assessments, reinforcing its utility for memory evaluation as noted by Gonçalves et al. (2017). However, unlike Collie, Maruff, Currie (2002), this study did not analyze differences between cognitively normal and impaired groups, which limits cross-group comparisons but confirms PAL's value alongside additional memory-related tests such as PRM and VRM. Furthermore, our findings resonate with those of Dassanayake, Ariyasinghe (2019) and Simpson et al. (2005), indicating that age and education are more influential than gender in determining cognitive performance. The superior processing speed observed among women in this study diverges from Simpson et al. (2005), potentially due to sample-specific age distributions. Lastly, in agreement with Bento-Torres et al. (2017), lower educational levels were associated with deficits in memory and switching/inhibition tasks; however, this study

uniquely identified age as a more significant factor than education in shaping performance. In summary, these findings reinforce the complementary nature of CANTAB and traditional tests while underscoring the necessity for culturally and demographically tailored cognitive assessment tools in understanding cognitive aging.

This study has several limitations that must be acknowledged. First, the sample size was relatively small (83 participants), with a disproportionately high representation of women (68.7%), and included only cognitively normal elderly individuals. These sample characteristics limit the generalizability of the findings and preclude the ability to differentiate between cognitive decline caused by normal aging and pathological conditions. Furthermore, the study did not rigorously control for demographic variables, which may have influenced the outcomes. Second, while the reliability of CANTAB has been extensively validated in international contexts (Backx et al., 2020; Gonçalves, Pinho, Simões 2015; Lowe, Rabbitt, 1998), it has not been verified in Korea, raising concerns about whether the test accurately reflects cognitive abilities without being affected by external factors. Third, the factor analysis relied solely on principal component analysis (PCA), which is limited in its ability to detect nonlinear relationships between variables and provides only exploratory insights. Future research should incorporate exploratory factor analysis (EFA) and confirmatory factor analysis (CFA) to validate the structure of cognitive constructs assessed by CANTAB. Fourth, the cross-sectional design of this study, which analyzed data at a single time point, does not allow for tracking changes in cognitive performance over time. Longitudinal studies are needed to explore how cognitive function evolves and to identify CANTAB indicators with high discriminative power for detecting age-related decline. Finally, the lack of a younger comparison group restricted this study's ability to assess age-related differences across broader age ranges.

To address these limitations, future research should focus on several key areas. Larger-scale studies with more diverse samples, including both cognitively normal and impaired individuals across a wide age range, are necessary to enhance generalizability and provide more comprehensive insights into cognitive aging. In particular, web-based experimental tools can facilitate the recruitment of participants across diverse age groups or large-scale samples, contributing to the generalizability of research findings (Tae, Choi, Lee, 2023). Conducting experiments with participants from various age ranges could further enhance the applicability of computerized neuropsychological tools such as CANTAB. Cultural adaptation and validation of CANTAB for use in Korea should be prioritized to ensure its cross-cultural applicability and measurement invariance. Advanced statistical techniques, such as structural equation modeling, should be employed to elucidate the complex relationships between cognitive domains and demographic factors. Additionally, recent studies have employed both logistic regression and decision tree analysis to examine cognitive impairments in the elderly, highlighting the importance of exploring the interactions between specific variables through these methods (Lee, Yang, 2024). Longitudinal investigations are crucial to explore the trajectory of cognitive

function and to identify early markers of cognitive decline. Additionally, research should examine the integration of CANTAB with other cognitive assessment tools and neuroimaging techniques to provide a more comprehensive evaluation of cognitive function in aging populations. By addressing these limitations and pursuing these research directions, future studies can refine the application of CANTAB, enhance its diagnostic utility, and support its adaptation to diverse demographic and cultural contexts, ultimately advancing its role in cognitive assessment for aging populations and contributing to early detection and intervention strategies for age-related cognitive decline.

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